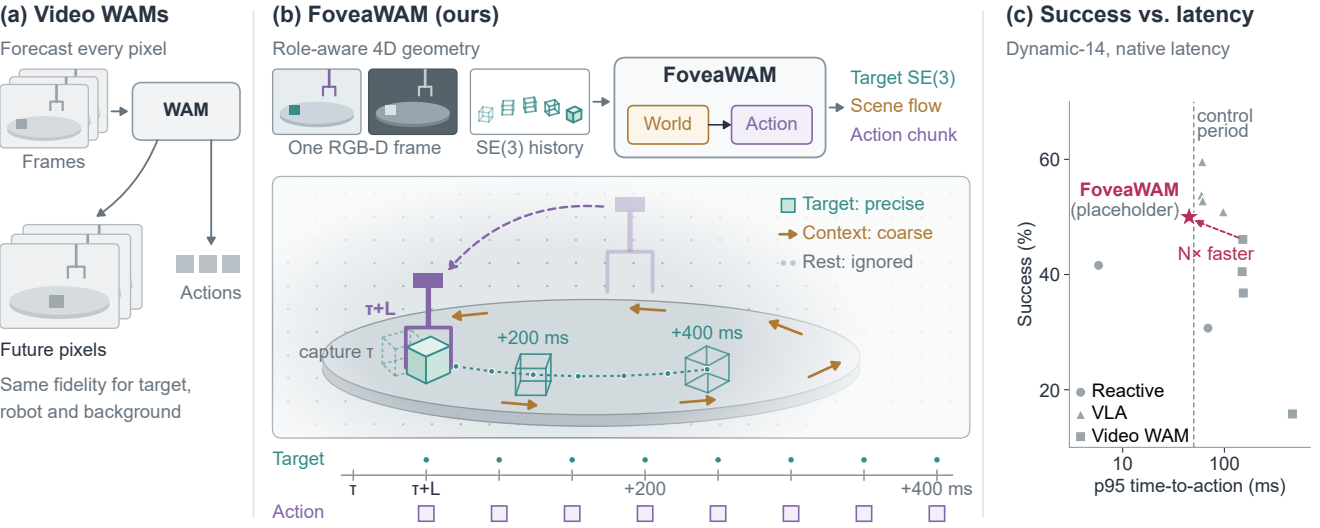

(a) Video WAMs
Forecast every pixel
WAM
Frames
Actions
Future pixels
Same fidelity for target, robot and background
(b) FoveaWAM (ours)
Role-aware 4D geometry
One RGB-D frame
SE(3) history
FoveaWAM
World
Action
Target SE(3)
Scene flow
Action chunk
Target: precise
Context: coarse
Rest: ignored
τ+L
capture τ
+200 ms
+400 ms
Target
Action
τ
τ+L
+200
+400 ms
(c) Success vs. latency
Dynamic-14, native latency
control period
FoveaWAM
(placeholder)
N× faster
Success (%)
60
40
20
Reactive
VLA
Video WAM
10
100
p95 time-to-action (ms)